

Symmetry-Matched Polarization Analysis in MnF_2 : Separating Spectral Resolution from Chiral Sensitivity in Altermagnetic Neutron Scattering

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Recent neutron-scattering studies of MnF_2 appear, at first sight, to pull in opposite directions: unpolarized inelastic neutron scattering reported no resolvable altermagnetic magnon splitting within experimental resolution, whereas later polarized measurements revealed a resolved magnon doublet and a polarization-odd chiral response. We show that these observations are not logically contradictory because spectral resolution and chiral sensitivity correspond to distinct measurement channels. Starting from the Blume–Maleev cross section, we decompose the transverse dynamical magnetic correlation tensor into polarization-even and polarization-odd parts. Reversing the incident neutron polarization isolates the antisymmetric (chiral) correlator without invoking coherent interference between distinct magnon final states. For the MnF_2 comparison considered here, we distinguish four experimentally relevant mechanisms that can produce a null: a physical null, a resolution null, a polarization-projection null, and antiferromagnetic-domain cancellation; this list is deliberately contextual rather than exhaustive. Applied to MnF_2 , this framework separates the approximately 0.2 meV magnon splitting reported in polarized HYSPEC data, which is dominated by long-range dipolar interactions, from the much smaller exchange-asymmetry scale $\delta J_7 = |J_{7a} - J_{7b}|$ that controls the altermagnetic chiral response. The reported 5–11% polarization-odd intensity therefore probes fine structure that is not equivalent to the dominant spectral splitting. We formulate a symmetry-matched observability criterion and an experimental protocol for testing chiral altermagnetic dynamics in MnF_2 and related compensated magnets.

I. INTRODUCTION

Altermagnets are collinear, compensated magnetic systems in which spin and crystal symmetries permit momentum-dependent spin splitting without a net ferromagnetic moment.[1, 2] Their ordered states can be described using symmetry frameworks that go beyond the conventional ferromagnet–antiferromagnet dichotomy, and their collective excitations may carry momentum-dependent chirality.[3, 5] In d -wave altermagnets, ferriocally ordered magnetic multipoles provide a useful order-parameter description; MnF_2 has played a central role in this development.[4]

MnF_2 is also a stringent experimental test because it has long served as a canonical collinear antiferromagnet. Conventional neutron spectroscopy successfully described its dominant spin-wave dispersion using a small set of exchange interactions and anisotropies.[13, 14] The modern altermagnetic question is therefore not whether MnF_2 is magnetically ordered, but which small symmetry-breaking features of its dynamical response are experimentally observable and which measurement channel carries them.

This question became particularly sharp after two recent neutron studies. Morano *et al.* reported no altermagnetic magnon-band splitting within the resolution of their unpolarized inelastic-neutron measurement and found the spectrum well described by an effectively higher-symmetry exchange model.[6] Faure *et al.* subsequently reported polarized inelastic-neutron data in which a magnon

splitting of order 0.2 meV is directly resolved, predominantly from long-range dipolar interactions, while the difference between opposite incident polarizations isolates a chiral response that reverses sign between the two magnon modes.[7] The latter polarization-odd contribution reaches approximately 5–11% of the total magnetic intensity depending on wave vector and is compatible with a much smaller symmetry-breaking exchange scale $\delta J_7 = |J_{7a} - J_{7b}|$ of only a few μeV . [7]

The purpose of this paper is to formalize why these measurements should not be treated as mutually exclusive verdicts on a single scalar observable. The central point is simple but consequential: *energy resolution and polarization sensitivity interrogate different mathematical components of the response*. A spectrometer can fail to resolve two nearby peaks even when the relevant excitation exists; separately, an unpolarized measurement can project out an antisymmetric correlation component even when the total magnetic intensity is large. These two null mechanisms must not be conflated.

We develop this distinction directly from the polarized neutron cross section. The resulting formulation removes three unnecessary assumptions: (i) it does not require the splitting to “cancel” in a sum of positive spectral weights; (ii) it does not require coherent interference between orthogonal magnon final states; and (iii) it does not claim that polarization analysis universally circumvents finite energy resolution. Instead, polarization reversal selects a different tensor component of the dynamical response. This leads naturally to a symmetry-matched observability condition and a practical experimental protocol.

The Blume–Maleev polarization formalism used below is established. The contribution of the present work is

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narrower: to organize resolution, polarization projection, and antiferromagnetic-domain averaging as distinct operations in the measurement chain, and to apply that separation consistently to the recent MnF_2 measurements. For the experimental comparison considered here, physical absence, finite spectral resolution, polarization projection, and domain cancellation are logically distinct null mechanisms. They are not asserted to exhaust every way a real experiment can yield a null.

II. MnF_2 AND THE HIERARCHY OF EXPERIMENTALLY RELEVANT SCALES

A. Conventional spin dynamics and altermagnetic symmetry

MnF_2 crystallizes in the rutile structure, space group $P4_2/mnm$ (No. 136), and orders antiferromagnetically below $T_N \approx 67$ K with the Mn moments aligned predominantly along the tetragonal c axis.[6, 7] Its dominant magnetic dynamics are governed by exchange interactions on the scale of tenths of meV, while the full magnon bandwidth is several meV. In contrast, symmetry-distinct longer-range exchange paths such as J_{7a} and J_{7b} can differ only at the μeV scale.[7]

Bhowal and Spaldin showed that the d -wave altermagnetic state of MnF_2 can be described by ferroically ordered magnetic octupoles, providing a multipolar language for the broken time-reversal symmetry and nonrelativistic spin splitting.[4] This order-parameter description should, however, be distinguished from the experimental observable measured in an inelastic neutron experiment. Polarized INS couples to dynamical magnetic correlation functions and, in the appropriate geometry, to their antisymmetric or chiral component. It does not by itself constitute a direct real-space measurement of a static magnetic multipole.

That distinction has become even more important in light of a recent polarized-neutron diffraction preprint that reconstructs the static magnetization density of MnF_2 and reports a symmetry-allowed rank-5 magnetic multipole component.[8] The present work therefore uses “altermagnetic chirality” for the INS observable and treats multipolar order as the symmetry context rather than identifying the INS asymmetry with a direct octupole measurement.

B. The Morano and Faure measurements ask different questions

Morano *et al.* used unpolarized INS to search for the expected splitting along selected high-symmetry directions. Within their experimental resolution they did not observe the predicted altermagnetic band separation, and the data were described by an exchange model whose leading interactions have a higher effective symmetry than the

full crystal structure.[6] This establishes a well-defined experimental statement: under that measurement configuration, no additional splitting was required by the data beyond the stated resolution and model sensitivity.

Faure *et al.* used HYSPEC in both unpolarized-equivalent and half-polarized configurations. Their data resolve a magnon doublet with a maximum splitting of approximately 0.2 meV, substantially larger than the few- μeV altermagnetic exchange-asymmetry scale. Their modeling attributes the dominant spectral splitting to long-range dipolar coupling. Crucially, the dipolar interaction by itself mixes magnon chiralities and does not generate the observed altermagnetic chiral response in the idealized higher-symmetry limit. Introducing a nonzero δJ_7 produces a finite polarization-odd term.[7]

The important hierarchy is therefore

$$\delta J_7 \sim \text{few } \mu\text{eV}, \quad (1)$$

$$\Delta E_{\text{dipolar}}^{\text{max}} \sim 0.2 \text{ meV}, \quad (2)$$

$$W_{\text{magnon}} \sim \text{several meV}, \quad (3)$$

with $\delta J_7 \ll \Delta E_{\text{dipolar}}^{\text{max}} \ll W_{\text{magnon}}$. The polarization experiment is scientifically valuable precisely because the parameter controlling the chiral response can be much smaller than the energy scale that dominates the observed doublet.

III. POLARIZED-NEUTRON RESPONSE AS A TENSOR PROJECTION

A. Transverse dynamical correlation tensor

Let $\mathbf{Q}$ and ω denote momentum and energy transfer. Define the transverse projector

$$\Pi_{\alpha\beta}(\mathbf{Q}) = \delta_{\alpha\beta} - \hat{Q}_\alpha \hat{Q}_\beta, \quad (4)$$

and the transverse magnetization operator

$$M_\alpha^\perp(\mathbf{Q}) = \Pi_{\alpha\beta}(\mathbf{Q}) M_\beta(\mathbf{Q}). \quad (5)$$

The corresponding dynamical correlation tensor is

$$S_{\alpha\beta}^\perp(\mathbf{Q}, \omega) = \frac{1}{2\pi\hbar} \int dt e^{-i\omega t} \langle M_\alpha^\perp(-\mathbf{Q}, 0) M_\beta^\perp(\mathbf{Q}, t) \rangle. \quad (6)$$

The exact normalization is convention dependent and is not important for the channel decomposition below.[9–11]

We separate Eq. (6) into symmetric and antisymmetric parts,

$$S^{(\alpha\beta)} = \frac{1}{2} (S^{\alpha\beta} + S^{\beta\alpha}), \quad (7)$$

$$S^{[\alpha\beta]} = \frac{1}{2} (S^{\alpha\beta} - S^{\beta\alpha}). \quad (8)$$

The polarization-even intensity is controlled by the symmetric contraction, whereas the polarization-odd chiral response is controlled by the antisymmetric part.

B. Unpolarized channel

For unpolarized incident neutrons, summing over the outgoing spin gives the familiar magnetic cross section

$$I_0(\mathbf{Q}, \omega) \propto \Pi_{\alpha\beta}(\mathbf{Q}) S_{\alpha\beta}(\mathbf{Q}, \omega). \quad (9)$$

Because $\Pi_{\alpha\beta}$ is symmetric, its contraction with $S^{[\alpha\beta]}$ vanishes identically. Thus Eq. (9) measures the polarization-even part of the response. This does *not* mean that two split magnon modes cancel from the unpolarized spectrum. If two modes have nonzero matrix elements at distinct energies, both contribute positive spectral weight and can, in principle, be resolved when the instrumental line shape is sufficiently narrow.

This distinction corrects a common but misleading shorthand. Unpolarized scattering can be insensitive to chirality while remaining sensitive to energy splitting. Whether the splitting is experimentally visible is then a separate resolution question.

C. Polarization-odd chiral channel

For a fixed incident polarization $\mathbf{P}_i$ without analyzing the outgoing neutron spin, the Blume–Maleev form used in the MnF_2 polarized experiment can be written schematically as [7, 10, 11]

$$I(\mathbf{P}_i) \propto \int dt e^{-i\omega t} [\langle \mathbf{M}_{-\mathbf{Q}}^\perp \cdot \mathbf{M}_{\mathbf{Q}}^\perp(t) \rangle + i\mathbf{P}_i \cdot \langle \mathbf{M}_{-\mathbf{Q}}^\perp \times \mathbf{M}_{\mathbf{Q}}^\perp(t) \rangle]. \quad (10)$$

The second term is odd under $\mathbf{P}_i \rightarrow -\mathbf{P}_i$. Although the cross product of complex magnetic amplitudes is imaginary in the relevant sense, the prefactor i makes its contribution to the measured intensity real.

Define

$$I_+(\mathbf{Q}, \omega) \equiv I(+\mathbf{P}_i), \quad (11)$$

$$I_-(\mathbf{Q}, \omega) \equiv I(-\mathbf{P}_i). \quad (12)$$

Then

$$I_+ + I_- = 2I_0, \quad (13)$$

$$I_+ - I_- = 2I_\chi, \quad (14)$$

where I_χ is the polarization-odd chiral contribution. Equations (13) and (14) are the essential measurement algebra: polarization reversal separates two response channels from the same sample.

Because $\mathbf{M}^\perp$ is perpendicular to $\mathbf{Q}$, the vector $\mathbf{M}_{-\mathbf{Q}}^\perp \times \mathbf{M}_{\mathbf{Q}}^\perp$ is parallel to $\hat{\mathbf{Q}}$. Consequently the chiral term carries a geometric factor proportional to

$$\mathbf{P}_i \cdot \hat{\mathbf{Q}}. \quad (15)$$

This is why aligning the incident polarization with the scattering vector is an efficient geometry for observing the chiral response. [7]

D. Mode-resolved interpretation without interbranch interference

For discrete one-magnon final states $|n\rangle$, the dynamical tensor has the spectral form

$$S_{\alpha\beta}^\perp(\mathbf{Q}, \omega) \propto \sum_n \langle 0 | M_\alpha^\perp(-\mathbf{Q}) | n \rangle \langle n | M_\beta^\perp(\mathbf{Q}) | 0 \rangle \delta(\omega - \omega_n). \quad (16)$$

The antisymmetric part of the *same mode's* matrix-element tensor measures the handedness of its precessional motion. This is the appropriate origin of mode-resolved magnon chirality. [12] Orthogonal final states are summed as probabilities; no coherent cross term between distinct magnon eigenstates is required.

This point matters for MnF_2 . The two observed magnon branches may carry opposite chiral signatures even though the experimental intensity is an incoherent sum over final states. The sign reversal is encoded in the antisymmetric matrix-element tensor of each mode, not in a phase-beating term between two different final states.

IV. RESOLUTION NULLS AND PROJECTION NULLS ARE MATHEMATICALLY DISTINCT

A. Instrumental resolution convolution

Let $R(\omega - \omega')$ be the normalized energy-resolution function. The measured intensities are convolutions of the intrinsic response,

$$I_\pm^{\text{obs}}(\omega) = \int d\omega' R(\omega - \omega') [I_0(\omega') \pm I_\chi(\omega')]. \quad (17)$$

Therefore

$$I_+^{\text{obs}} + I_-^{\text{obs}} = 2 R * I_0, \quad (18)$$

$$I_+^{\text{obs}} - I_-^{\text{obs}} = 2 R * I_\chi. \quad (19)$$

A resolution-limited doublet can merge into one broad feature in $R * I_0$ even when two intrinsic poles are present. This can be made quantitative in a minimal Gaussian model. Suppose two equal-weight lines are centered at $\omega = \pm\Delta/2$ and, after combining intrinsic and instrumental Gaussian widths in quadrature, each has standard deviation σ_{eff} . Then

$$I(\omega) \propto \exp\left[-\frac{(\omega - \Delta/2)^2}{2\sigma_{\text{eff}}^2}\right] + \exp\left[-\frac{(\omega + \Delta/2)^2}{2\sigma_{\text{eff}}^2}\right]. \quad (20)$$

The curvature at the midpoint changes sign at

$$\Delta = 2\sigma_{\text{eff}} = \frac{\Gamma_{\text{eff}}}{\sqrt{2 \ln 2}} \approx 0.849 \Gamma_{\text{eff}}, \quad (21)$$

where $\Gamma_{\text{eff}} = 2\sqrt{2 \ln 2} \sigma_{\text{eff}}$ is the single-line FWHM. Thus, for two equal Gaussian components, $\Delta/\Gamma_{\text{eff}} < 0.849$ produces a single central maximum, whereas a central dip

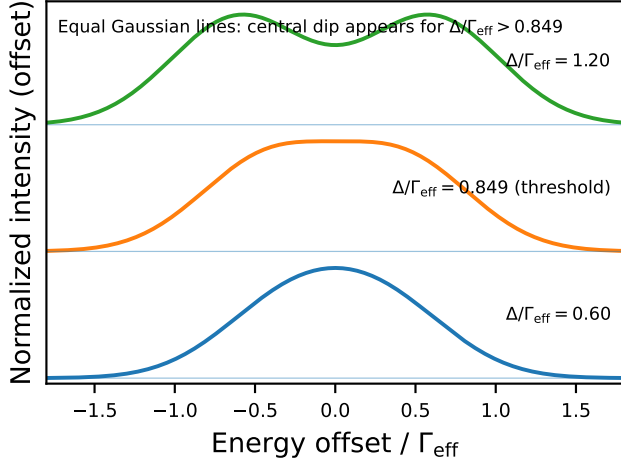


FIG. 1. Quantitative Gaussian illustration of a resolution null. The horizontal coordinate is energy offset in units of the effective single-line FWHM Γ_{eff} . Two equal Gaussian components are shown for three ratios $\Delta/\Gamma_{\text{eff}}$; the central curvature changes sign at the analytic threshold $1/\sqrt{2\ln 2} \simeq 0.849$. The construction is dimensionless and is not a fit to either MnF_2 experiment. Real neutron resolution is generally multidimensional and need not be Gaussian. The figure was produced with deterministic plotting code developed with substantive assistance from OpenAI ChatGPT (GPT-5.6 Sol); the equations, parameter ratios, and interpretation were specified and verified by the author.

appears above that threshold. Equation (21) is not a universal instrumental resolution criterion: unequal spectral weights, non-Gaussian four-dimensional resolution, dispersion within the integration volume, background, and fitting strategy all shift practical resolvability. It is included only to demonstrate analytically that finite positive spectral weights can be present while a doublet remains unresolved. Figure 1 illustrates the dimensionless crossover.

Polarization analysis does not abolish Eq. (17). If two opposite-chirality contributions overlap perfectly in energy with equal magnitude, the convolved difference can also cancel. The advantage of polarization reversal is instead that it projects onto I_χ rather than I_0 . This can expose small microscopic parameters through intensity contrast even when those parameters make only a tiny contribution to the energy eigenvalues.

B. Projection null

A polarization-projection null occurs when the physical chiral correlator is nonzero but the measurement functional annihilates it. Examples include an unpolarized incident beam, averaging equally over opposite incident polarizations, or choosing $\mathbf{P}_i \perp \mathbf{Q}$ in the half-polarized geometry of Eq. (10). In all such cases, a null chiral channel does not imply $S^{[\alpha\beta]} = 0$.

C. Domain cancellation

Time-reversed antiferromagnetic domains produce opposite chiral signals. Let their populations be p_+ and p_- , normalized so $p_+ + p_- = 1$. To leading order the measured chiral term is diluted by

$$\eta_D = p_+ - p_-, \quad I_\chi^{\text{obs}} = \eta_D I_\chi^{\text{single domain}}. \quad (22)$$

Using the roughly 4:1 domain-population estimate reported by Faure *et al.* as an illustrative value gives

$$\eta_D = \frac{4 - 1}{4 + 1} = 0.6. \quad (23)$$

This value should be understood as an approximate domain-dilution estimate rather than a precision determination of η_D . [7] An exactly balanced multidomain sample would have $\eta_D = 0$ and could therefore show a vanishing polarization-odd signal despite locally chiral excitations.

V. A SYMMETRY-MATCHED OBSERVABILITY CONDITION

The previous sections suggest a compact operational criterion. Define the scalar chiral spectral density

$$\mathcal{C}(\mathbf{Q}, \omega) \equiv i\hat{\mathbf{Q}} \cdot \int dt e^{-i\omega t} \langle \mathbf{M}_{-\mathbf{Q}}^\perp \times \mathbf{M}_{\mathbf{Q}}^\perp(t) \rangle. \quad (24)$$

For the half-polarized geometry, the polarization-odd observable is proportional to

$$\mathcal{O}_\chi(\mathbf{Q}, \omega) \propto \eta_D \left(\mathbf{P}_i \cdot \hat{\mathbf{Q}} \right) \mathcal{C}(\mathbf{Q}, \omega), \quad (25)$$

followed by convolution with instrumental resolution and acceptance functions.

Equation (25) is the symmetry-matched polarization condition in operational form. A nonzero expectation value in the polarization-odd measurement channel requires all of the following:

1. a nonzero antisymmetric magnetic correlator $\mathcal{C}$ at the chosen $(\mathbf{Q}, \omega)$;
2. a probe polarization with nonzero projection onto the allowed chiral vector, here $\mathbf{P}_i \cdot \hat{\mathbf{Q}} \neq 0$; and
3. a nonzero net domain imbalance η_D unless individual domains are spatially resolved.

Experimental detection of that contribution additionally requires adequate instrumental acceptance and statistical sensitivity after resolution convolution.

This formulation avoids assigning the neutron spin to an incorrect polar-vector irreducible representation. Neutron polarization is an axial vector. The crystal-specific symmetry information enters through the allowed tensor structure of $\mathcal{C}(\mathbf{Q}, \omega)$ and through the momentum directions on which it must vanish or change sign. The probe requirement is then expressed as a nonzero contraction with that allowed tensor component.

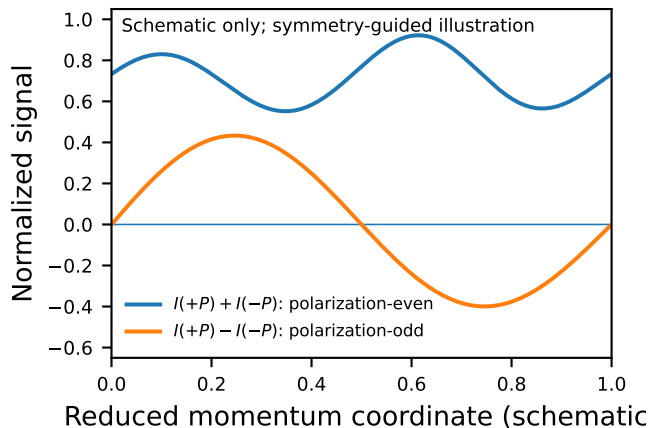


FIG. 2. Schematic separation of the polarization-even sum and polarization-odd difference channels under incident-polarization reversal. The difference channel isolates the antisymmetric magnetic correlator. The illustration is symmetry guided but is not experimental data. The schematic was produced with deterministic plotting code developed with substantive assistance from OpenAI ChatGPT (GPT-5.6 Sol); the channel definitions and displayed relationships were specified and verified by the author.

VI. APPLICATION TO MnF_2

A. What the unpolarized null does and does not establish

The Morano *et al.* result establishes that their unpolarized spectrum did not require the predicted altermagnetic magnon splitting within the resolution and uncertainty of that experiment.[6] It does not establish that every symmetry-allowed chiral correlator is zero, because the unpolarized cross section in Eq. (9) removes the antisymmetric response by construction.

Conversely, a sufficiently high-resolution unpolarized experiment can resolve two modes if they carry nonzero symmetric spectral weight and are separated by more than the effective line width. Therefore the limitation of unpolarized scattering is not absolute with respect to *energy splitting*; it is absolute only with respect to the specific polarization-odd observable discarded by the spin average.

B. What the polarized measurement adds

The Faure *et al.* measurement adds two logically distinct observations. First, the polarization-even sum resolves a magnon splitting of approximately 0.2 meV, which their modeling attributes predominantly to long-range dipolar coupling.[7] Second, the polarization-odd difference exhibits a sign-changing magnetic signal associated with the two magnon branches. The integrated chiral

contribution is reported to be approximately 5–11% of the total intensity depending on wave vector.[7]

The reported exchange parameters illustrate why these channels should not be equated. A representative fit uses $J_2 \approx 0.287$ meV and values of J_{7a} and J_{7b} differing by only a few μeV ; the polarization analysis constrains δJ_7 to this fine scale even though the visible doublet is dominated by a much larger dipolar contribution.[7] In other words,

$$\Delta E_{\text{dominant}} \neq E_{\text{chiral scale}}. \quad (26)$$

This is the central reconciliation.

C. The Morano–Faure comparison as complementarity

It is therefore preferable to describe the two experiments as complementary rather than contradictory. Morano *et al.* constrain the symmetric spectral response under one instrumental resolution and model class. Faure *et al.* resolve a larger dipolar splitting and additionally measure an antisymmetric polarization-odd response that is sensitive to a much smaller altermagnetic parameter. The later result does not require the earlier experiment to have “missed” a 0.2 meV feature under identical conditions; the experiments used different instruments, samples, configurations, and analysis channels.

A robust comparison should therefore report at least four quantities separately: (i) the energy-resolution function, (ii) the observed polarization-even dispersion, (iii) the polarization-odd chiral intensity, and (iv) the antiferromagnetic-domain population. Folding them into a single yes/no statement about “the splitting” obscures the physics.

VII. EXPERIMENTAL PROTOCOL FOR SYMMETRY-MATCHED DETECTION

The formalism suggests a minimal protocol for testing chiral altermagnetic dynamics in MnF_2 or a related compensated magnet.

1. **Establish the domain population.** Use polarized diffraction or an equivalent domain-sensitive measurement to determine p_+ and p_- . A domain-biased sample is strongly preferred.
2. **Choose symmetry-informative momentum trajectories.** In MnF_2 , the polarized HYSPEC work identifies the $[HH1]$ direction as useful, with the chiral response exhibiting characteristic momentum dependence and sign constraints.[7]
3. **Align the incident polarization to maximize the chiral contraction.** For the half-polarized geometry, choose $\mathbf{P}_i$ with a large component parallel

to $\mathbf{Q}$ over the energy and momentum window of interest.

4. **Acquire matched $+\mathbf{P}_i$ and $-\mathbf{P}_i$ data.** All other instrumental settings should remain fixed as closely as possible.
5. **Construct sum and difference channels.**

$$I_{\text{even}} = \frac{I_+ + I_-}{2}, \quad (27)$$

$$I_{\text{odd}} = \frac{I_+ - I_-}{2}, \quad (28)$$

$$A_\chi = \frac{I_+ - I_-}{I_+ + I_-}. \quad (29)$$

The sum probes conventional magnetic spectral weight; the difference isolates the chiral correlator; the normalized asymmetry is useful for comparing momentum points with different total intensity.

6. **Fit both channels with the same Hamiltonian.** A valid microscopic model should reproduce the dispersion and the polarization-odd response simultaneously. A model that fits only the energy eigenvalues can possess a spurious effective symmetry and still fail the chiral test.
7. **Convolve the model with the measured resolution function.** Claims of absence or detectability should be made only after applying the actual four-dimensional instrumental resolution and experimental integration windows.
8. **Correct or report domain dilution.** Use Eq. (22) to distinguish intrinsic single-domain chirality from the measured multidomain asymmetry.

This protocol does not require full outgoing-polarization analysis in every implementation. The Faure experiment demonstrates that a half-polarized configuration with controlled incident polarization and summed outgoing spin can isolate the required term through polarization reversal.[7]

VIII. FALSIFIABLE CHECKS AND DIAGNOSTIC NULL TESTS

A useful symmetry-matched framework should generate null tests as well as positive predictions. For the geometry described above, the following checks are particularly diagnostic.

1. **Polarization reversal:** I_{odd} must change sign under $\mathbf{P}_i \rightarrow -\mathbf{P}_i$, while I_{even} is invariant within systematic uncertainties.
2. **Polarization rotation:** rotating $\mathbf{P}_i$ toward a direction perpendicular to $\mathbf{Q}$ should suppress the chiral term according to Eq. (15).

3. **Domain reversal:** reversing the dominant antiferromagnetic domain should reverse the sign of I_χ at fixed $(\mathbf{Q}, \omega)$, while leaving the leading polarization-even spectrum essentially unchanged.

4. **Balanced-domain limit:** preparing equal populations of time-reversed domains should suppress the net chiral difference even if each domain separately supports chiral magnons.

5. **Symmetry-related wave vectors:** the chiral signal should obey the sign changes and nodes required by the magnetocrystalline symmetry. Faure *et al.* report such sign structure for symmetry-related directions and a characteristic reduction of the chiral contribution near the relevant zone boundary.[7]

6. **Resolution scaling:** improving energy resolution should sharpen I_{even} and may resolve additional spectral structure, but it should not create a polarization-odd signal when $\mathcal{C} = 0$ or when the polarization/domain factors in Eq. (25) vanish.

These tests distinguish a true physical absence of chirality from a null caused by the probe, domains, or resolution.

IX. IMPLICATIONS FOR MAGNETIC MULTIPOLES AND HIDDEN ORDER

The term “hidden order” is often used loosely for magnetic quantities that produce weak or symmetry-selective experimental signatures. In MnF_2 , several distinct statements must be kept separate.

First, magnetic octupoles provide a symmetry-based order-parameter description of d -wave altermagnetism and are directly connected to nonrelativistic spin splitting in the theoretical framework of Bhowal and Spaldin.[4] Second, polarized INS measures dynamical spin correlations and can identify the chirality of magnon branches.[5, 7] Third, a recent polarized-diffraction preprint reconstructs static real-space magnetization density and reports higher-rank symmetry-allowed multipolar structure in MnF_2 . [8]

These are complementary layers of description, not interchangeable measurements. The present analysis therefore supports a narrower but stronger conclusion: *symmetry-matched polarization analysis can reveal dynamical correlation components that are projected out by an unpolarized measurement.* It does not claim that every polarization-odd INS signal is a direct measurement of a specific static multipole rank.

This distinction is experimentally useful because it suggests a multimodal strategy. Polarized diffraction constrains static domain and multipolar structure; polarized INS constrains branch-resolved chirality; high-resolution unpolarized INS constrains the polarization-even dispersion; and electronic probes constrain nonrelativistic band splitting. Agreement among these observables provides a much stronger identification of altermagnetic order than any single spectral feature.

X. A CONTEXTUAL FOUR-NULL TAXONOMY FOR THE MnF_2 COMPARISON

The MnF_2 comparison is usefully organized by four null mechanisms that arise directly in the measurement chain discussed above:

Within the stated model and target observable, only the first null directly establishes the absence of the targeted correlation function. The other three are statements about the measurement map from the physical state to the recorded observable. The table is not intended as an exhaustive taxonomy of experimental nulls: background or normalization errors, nuisance-parameter confounding, sample-state drift, unmodeled acceptance effects, and model misspecification can also generate or mimic null results. The four entries are singled out because they are the mechanisms required to separate the resolution and polarization questions at issue in the MnF_2 measurements.

XI. SCOPE AND LIMITATIONS

This paper is a theoretical and methodological analysis of published neutron-scattering results; it reports no new neutron measurements and performs no independent refit of raw experimental data. Numerical values quoted for MnF_2 are taken from the cited literature. The Gaussian calculation in Eqs. (20)–(21) is an analytic illustration of resolvability, not a surrogate resolution model for either experiment. A definitive quantitative comparison between experiments would require access to the full four-dimensional resolution functions, integration windows, raw or reduced data, background treatment, sample mosaic, domain populations, and a common Hamiltonian parameterization.

The symmetry-matched criterion in Eq. (25) is deliberately operational. It captures the half-polarized geometry relevant to the published MnF_2 experiment but does not replace the full Blume–Maleev polarization-transfer formalism when outgoing polarization is analyzed or when nuclear–magnetic interference is important. Similarly, the multipolar interpretation of MnF_2 continues to develop; the present argument depends only on the existence of a nonzero antisymmetric dynamical correlation function and not on assigning a unique static multipole rank to that response.

TABLE I. Four null mechanisms relevant to the present MnF_2 comparison. The list is contextual, not exhaustive.

Null type	Cause	Meaning
Physical	$\mathcal{C} = 0$	Correlation absent
Resolution	$R * I$ merges features	Unresolved structure
Projection	contraction = 0	Channel projected out
Domain	$\eta_D = 0$	Domains cancel

XII. CONCLUSION

The recent MnF_2 neutron results are best understood by separating two questions that are often collapsed into one: *Can the experiment resolve two excitation energies?* and *Does the experiment measure the antisymmetric correlation channel that carries magnon chirality?*

Unpolarized INS measures the polarization-even contraction of the transverse dynamical structure factor. It can resolve a magnon doublet when the spectral separation exceeds the effective instrumental linewidth, but it discards the polarization-odd chiral correlator. Reversing an incident neutron polarization adds a second measurement channel whose difference isolates that antisymmetric response. That difference channel can provide branch-resolved signed contrast when the contributing modes remain distinguishable after instrumental convolution, and it requires no coherent interference between distinct magnon final states.

In MnF_2 , this distinction separates a resolved approximately 0.2 meV splitting dominated by dipolar interactions from a much smaller δJ_7 scale that controls the altermagnetic chiral response. The 5–11% polarization-odd intensity reported by Faure *et al.* is therefore not simply a sub-resolution measurement of the same scalar splitting sought by Morano *et al.*; it is a different observable with different symmetry and domain requirements.

For the class of polarized-neutron measurements analyzed here, the broader lesson is captured by Eq. (25): observability depends jointly on the physical correlation, probe geometry, domain population, and instrumental transfer function. A null result in one projected channel should not be promoted to a statement of physical absence without testing the relevant measurement-map alternatives and ordinary experimental systematics. In MnF_2 , symmetry-matched polarization analysis provides a direct and falsifiable way to separate the resolution, projection, and domain mechanisms that are central to the published comparison.

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DATA AVAILABILITY

This theoretical and methodological study generated no new experimental datasets and did not reanalyze raw neutron-scattering data. All numerical values and experimental observations discussed are taken from the cited

public literature and associated public records.

CONFLICT OF INTEREST

The author declares no competing financial interest related to this manuscript.

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